



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2022

Paper: Organic Chemistry

Course Code: CHEM-317

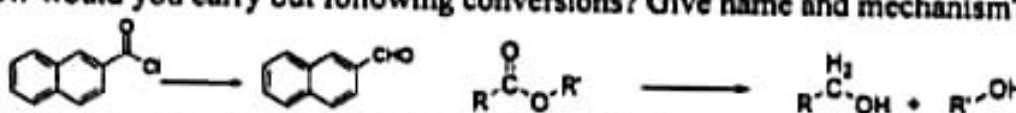
Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

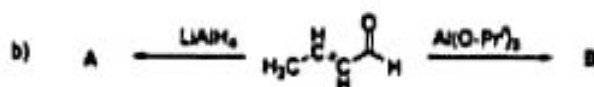
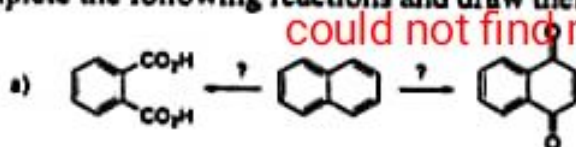
(6x5=30)

- Explain Oppenauer oxidation give mechanism and describe its role in oxidation of secondary alcohols?
- How would you carry out following conversions? Give name and mechanism?

- Describe Perch method for detection of free radicals.
- Describe instrumentation and sample handling in UV/VIS spectrophotometer.
- Describe Clemmensen and Wolf-Kishner reduction with mechanism for aldehydes and ketones.
- Define Lambert Beer law and Differentiate hypochromic and bathochromic shift?

Answer the following questions.

- Q. No. 2**
- Give mechanism of Corey-Kim oxidation of primary alcohols to aldehyde. (5)
 - What is reductive amination? Give different variations and modifications of reaction. (5)

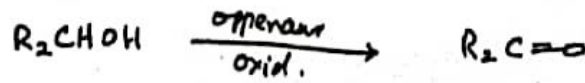
Q. No. 3 Complete the following reactions and draw their mechanisms: (2 x 5 = 10)



- Q. No. 4**
- Write a note on applications of infrared spectroscopy. (5)
 - Explain the method for the selective oxidation of ArCH_3 . (5)

(ii) Oppenauer Oxidation : "The method which use Aluminium t-butoxide (as base) and large excess of acetone to oxidized a secondary alcohol is called oppenauer oxidation."

- Acetone acts as hydrogen acceptor within a cyclic complex

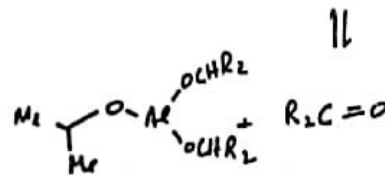
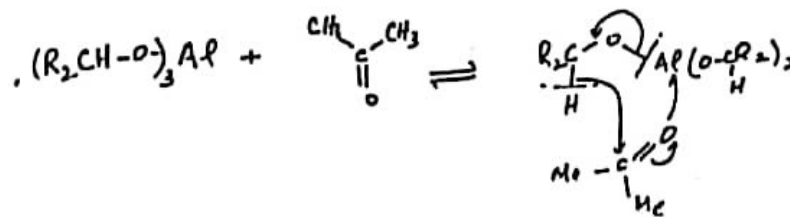
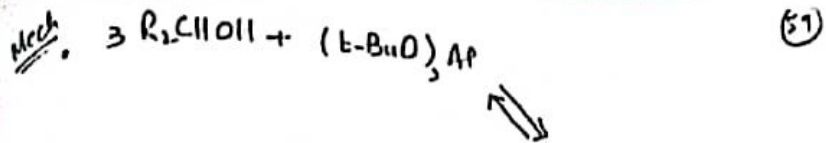


Question # (1)

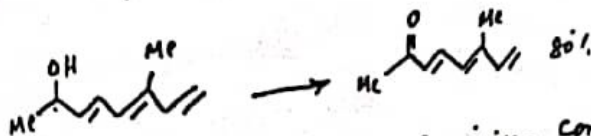
P.T.O

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- Benzene and toluene is often added a co-solvent to raise the temperature or cyclohexanone is used instead of acetone.
- It is specific for alcohol and thus suitable for compounds containing $C=C$ or phenolic groups e.g.



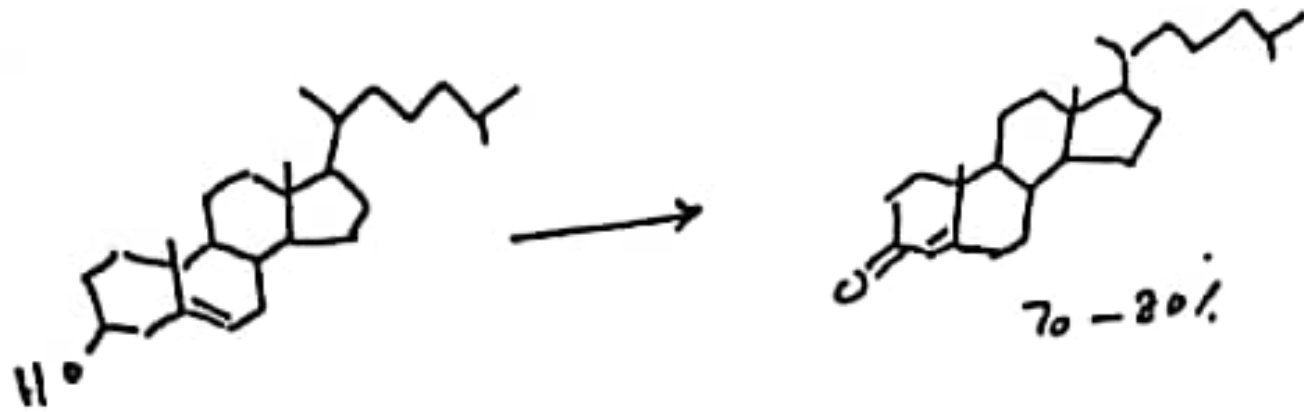
- (iii) In this method, Aluminium compounds are used which are basic in nature and can

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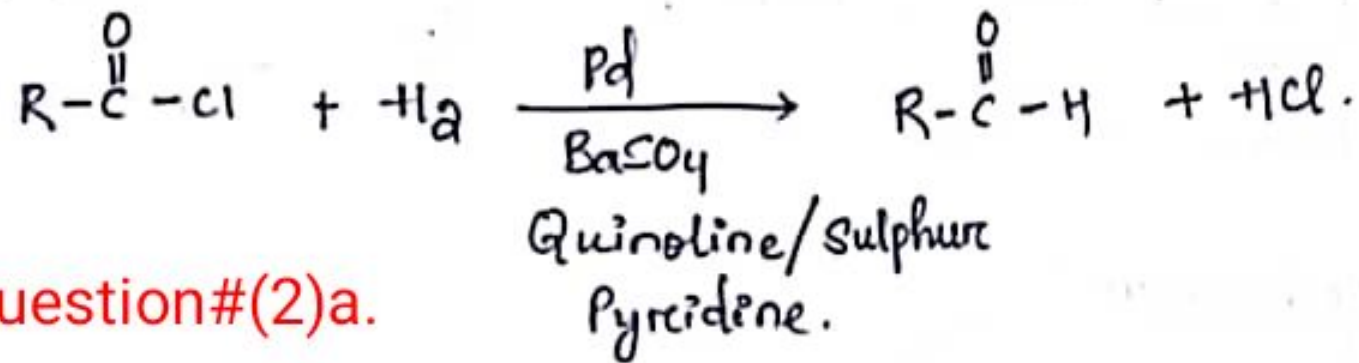
bring about prototropic shift within the molecule e.g. oxidation of cholesterol is accompanied by migration of $C=C$ to give $\alpha\beta$ -unsaturated ketone.



(b), Aldehydes from primary alcohol cannot be prepared because basic medium induces reaction b/w aldehyde and ketone used.

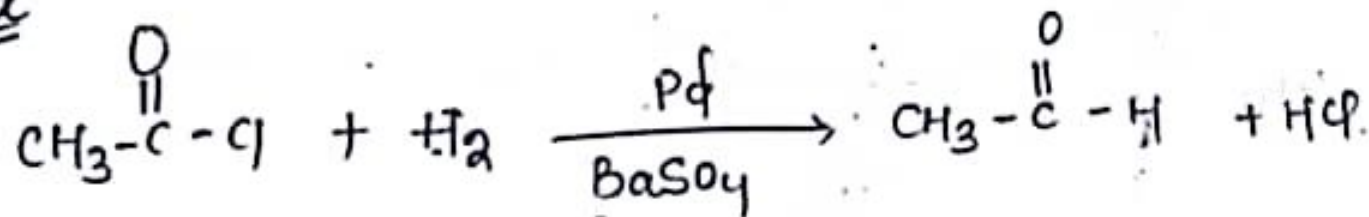
Overall, the Oppenauer oxidation provides a relatively mild and selective method for converting secondary alcohols to ketones, avoiding over-oxidation to carboxylic acids, which can occur in more aggressive oxidation conditions. This reaction is particularly useful in organic synthesis for the preparation of ketones.

① Rosenmund's Reaction :- (from acid chloride)

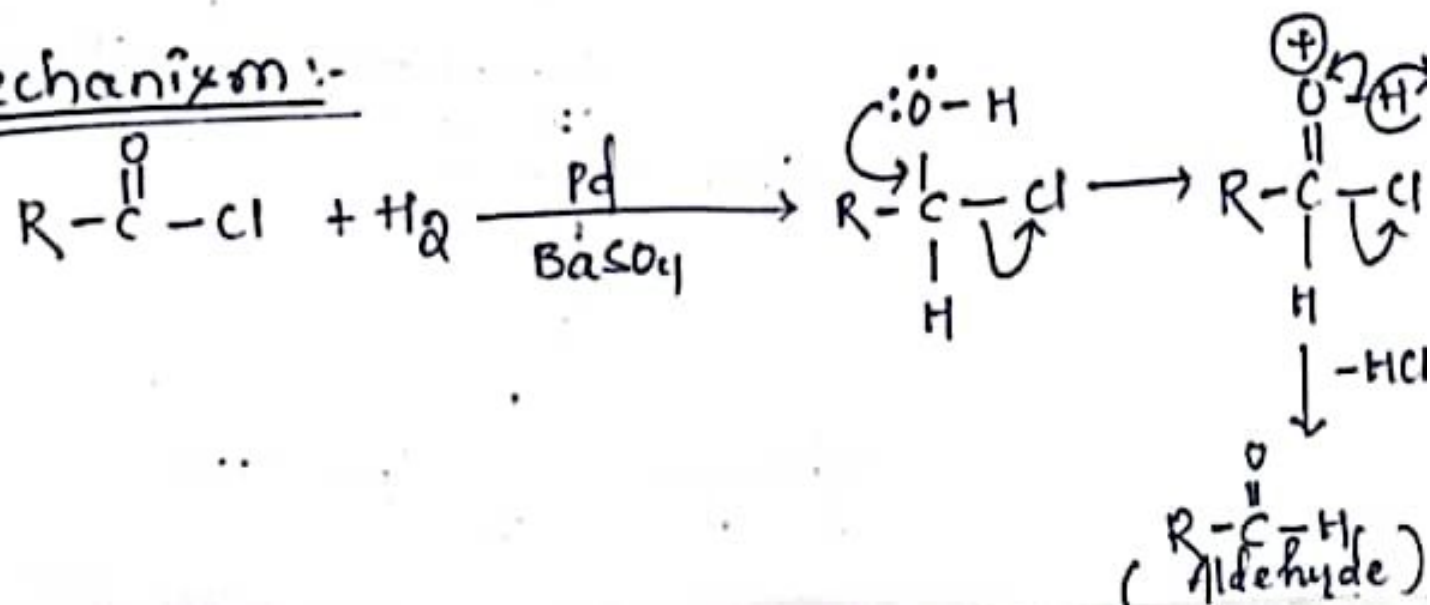


Question#(2)a.

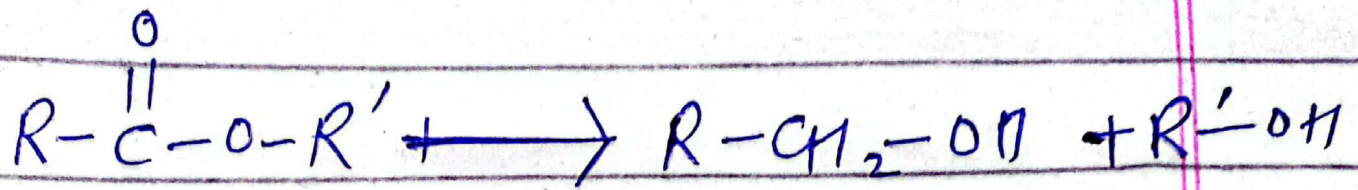
Ex



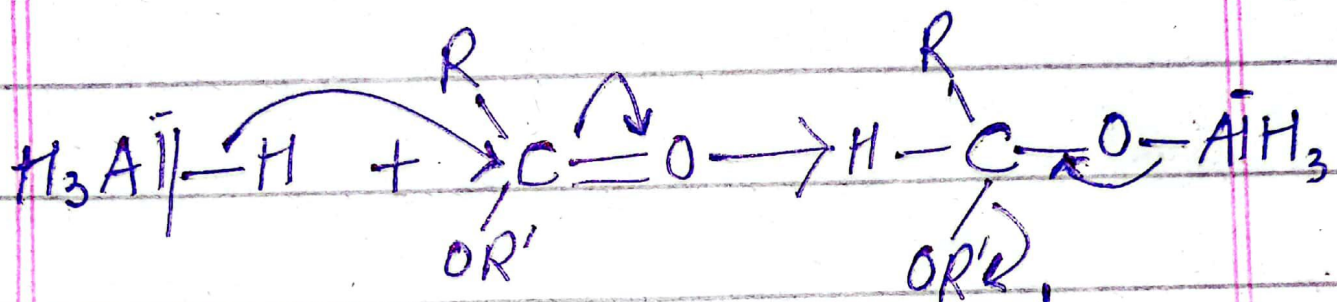
Mechanism:-



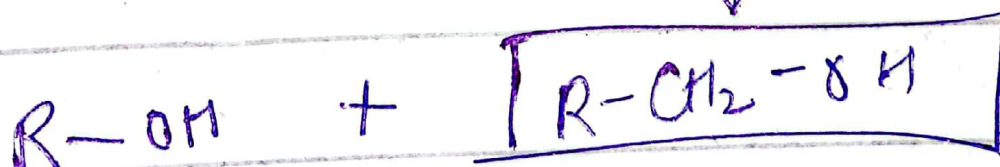
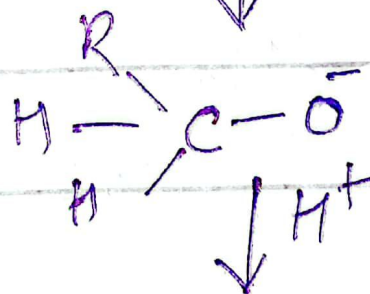
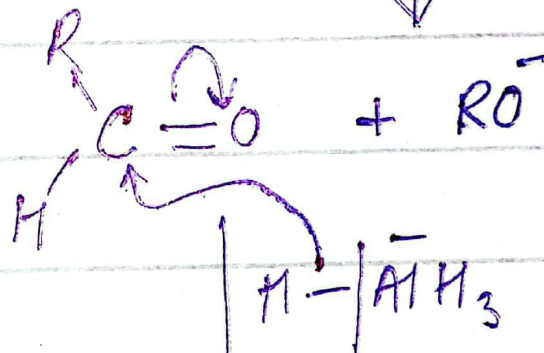
Ester to Alcohol conversion



$LiAlH_4$ reduces carboxylic acids and their all derivatives to either alcohol or amine.



Question #(2)b.





The Perch method, or the Photoinduced Electron Spin Resonance (ESR) by Continuous Wave Irradiation (CW), is a technique used for the detection of free radicals. Here's a basic description of the method: **Question#(3)**

1. **Sample Preparation:** To start, a sample containing the free radicals of interest is prepared. This could be a solution, solid material, or biological tissue.
2. **ESR Spectroscopy:** The Perch method relies on Electron Spin Resonance spectroscopy. ESR is a technique that detects unpaired electrons, such as those found in free radicals. These unpaired electrons have a magnetic moment





electrons have a magnetic moment and interact with an external magnetic field.

3. Irradiation: In the Perch method, the sample is continuously irradiated with microwave radiation at a specific frequency that matches the energy gap between the two electron spin states. This radiation induces transitions of electrons between these spin states.
4. Signal Detection: ESR detects changes in the magnetic resonance of the free radicals when they absorb energy and transition between spin states. This change in magnetic resonance is recorded as an ESR signal.
5. Continuous Wave Irradiation: Unlike  some other ESR techniques that use

some other ESR techniques that use pulsed microwave irradiation, the Perch method employs continuous wave irradiation. This means that the microwave radiation is applied continuously, allowing for the observation of steady-state ESR signals.

6. **Signal Analysis:** The ESR signal is analyzed to determine the concentration of free radicals in the sample and obtain information about their properties, such as their g -factor (a measure of the magnetic field at which the resonance occurs) and hyperfine coupling constants.

Overall, the Perch method is a valuable tool for studying free radicals and their behavior in various



samples, providing insights into their



6. **Signal Analysis:** The ESR signal is analyzed to determine the concentration of free radicals in the sample and obtain information about their properties, such as their g-factor (a measure of the magnetic field at which the resonance occurs) and hyperfine coupling constants.

Overall, the Perch method is a valuable tool for studying free radicals and their behavior in various samples, providing insights into their reactivity and properties. It's widely used in chemistry, biology, and materials science to investigate free radical species.



2.4 INSTRUMENTATION FOR UV-VIS SPECTROMETRY

Today a wide range of instruments are available for making molecular absorption measurements in the UV-visible range. These vary from simple and inexpensive machines for routine work to highly sophisticated devices that can be used for specialised work and of course the routine jobs also. However, the basic components of these instruments remain the same. The five essential components of UV-VIS instruments are as follows

- A stable radiation source
- Wavelength selector
- Sample holder
- Radiation detector or transducer, and
- Signal processing and output device

Question#(4)

The general layout of the essential components in a simple absorption instrument is given in Fig. 2.10.

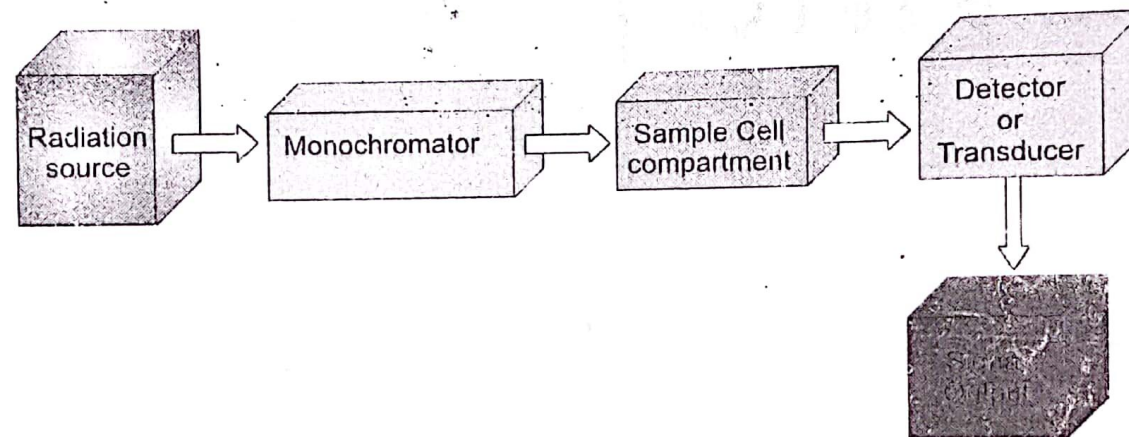


Fig. 2.10: General layout of the essential components in a simple absorption instrument

Let us learn about these components one by one.

2.4.1 Radiation Sources

A spectrophotometric radiation source must provide a stable high energy output over a broad range of wavelengths. There is no inexpensive source available that may provide stable output over the entire UV-visible range (190 nm to 780 nm). The radiation sources commonly used for the UV and visible range are described below.

Sources for UV Region

For measurements in the UV region, electric discharge sources like hydrogen or a deuterium lamp are used. In these, the excitation of the gaseous molecules is brought about by the passage of electrons through the gas at low pressures. A hydrogen lamp is commonly used in the spectrophotometers and gives light in the wavelength region of 160-375 nm. The radiant power of the hydrogen lamp is low and these are replaced by deuterium lamps but it increases the cost of the instrument.

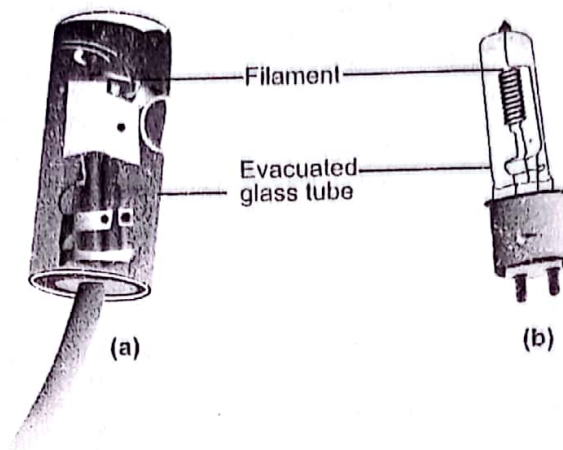


Fig. 2.11: Radiation sources: a) deuterium lamp for UV range b) tungsten lamp for visible range

Sources for Visible Region

In the beginning of development of instruments for colorimetric work, sunlight was used as source for measurements in the visible range. The modern instruments use a tungsten filament lamp as the radiation source. This consists of a thin, coiled tungsten wire that is sealed in an evacuated glass bulb. This gives radiations in the range of 350-2200 nm. As the output depends on the voltage, the tungsten lamp is energised by a 6 or 12 volt storage battery or by the output of a constant voltage transformer.

Now a days some instruments use tungsten-halogen lamps that contain a small amount of iodine in the quartz bulb housing the tungsten filament. The presence of iodine extends the output wavelength range of the lamp from 240 -2500 nm.

2.4.2 Wavelength Selectors

In spectrophotometric measurements we need to use a narrow band of wavelengths of light. This enhances the selectivity and sensitivity of the instrument. Less expensive instruments use a filter to isolate the radiant energy and provide a broad band of the wavelengths. In many applications we need to continuously vary the wavelength over a defined range. This can be achieved by using monochromators. Most modern instruments use monochromators that employ a prism or diffraction grating as the dispersing medium. The selection and use of filters and monochromators is described below.

Absorption Filters

In low cost instruments catering to measurements in the visible range, coloured glass

filters are used to cut off undesirable wavelengths. A typical filter is a coloured piece of glass, which absorbs light of certain wavelength and allows that of the other to pass through. You know that white light is made up of seven different colours; the acronym is **VIBGYOR** for violet, indigo, blue, green, yellow, orange and red. These seven colours add up to give white light back. When white light falls on an object, a part of it is absorbed and rest is transmitted. These transmitted components add up to give the observed colour of the object. The absorbed component and the observed colour can again add up to give white light. These are therefore called as **complementary** to each other. An object of a particular colour looks of that colour because this colour is transmitted and its complementary colour is absorbed.

In order to select a suitable filter for a measurement of absorbance of a solution, we need to have an understanding of complementary colours. The colour of the filter should be the complementary to colour of the solution to be measured. If a solution appears orange, this implies that orange light is not being absorbed but that the complement to orange i.e. blue is being absorbed. Therefore, for measuring absorbance of an orange solution you would employ a blue filter. The use of such a filter is illustrated in Fig. 2.12.

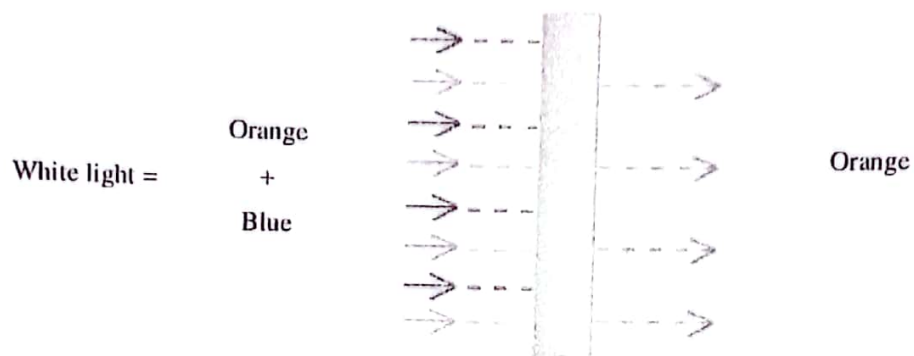


Fig. 2.12: Use of an absorption filter for the measurement of absorbance

Since these filters work by absorbing part of the radiation incident on them therefore they do not provide monochromatic light. In order to get a narrow band of wavelengths we can use certain filters which actually contain two glass filters: one filter absorbs strongly above a certain wavelength, while the second absorb strongly below a certain wavelength.

In order to find the colour of the filter to be used we can take the help of colour wheel shown in Fig. 2.13, or consult a table of complementary colours given in Table 2.2.

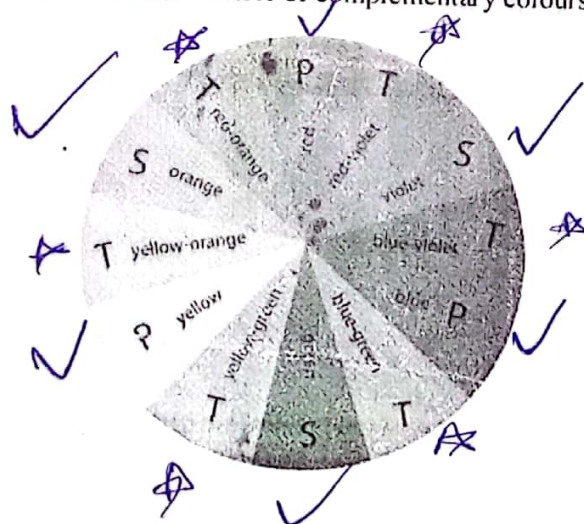


Fig. 2.13: The colour wheel

In a colour wheel, the colours, which face one another, are said to be complementary to each other. In order to measure a red coloured solution, its complementary, that is, a green coloured filter should be used. In the colour wheel P, S and T refer to the primary, secondary and tertiary colours, respectively.

Table 2.2: Complementary Colours Chart

Wavelength region eliminated by absorption (nm)	Colour absorbed	Complementary colour of the residual light (as seen by the eye)
400-450	violet	yellow-green
450-480	blue	yellow
480-490	green-blue	orange
490-500	blue-green	red
500-560	green	purple
560-580	yellow-green	violet
580-600	yellow	blue
600-650	orange	green-blue
650-750	red	blue-green

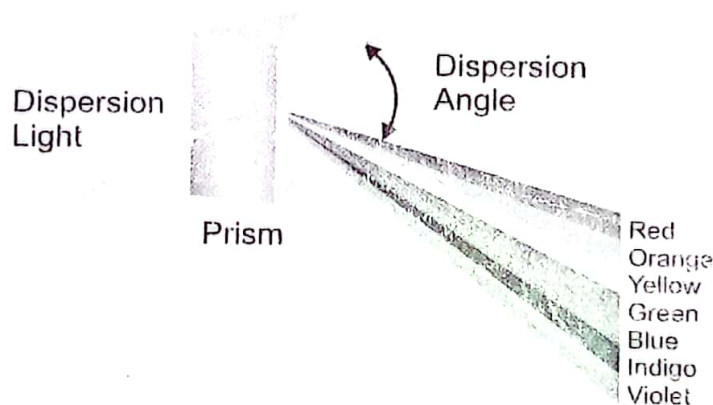
Absorption filters are simple and are totally adequate for many applications in visible range. However, for extended ranges we need **interference filters**. The interference filters cover a wider range than the absorption filters. Interference filters are essentially composed of two transparent parallel films of silver, which are so close as to produce interference effects. Such interference filters are available for ultraviolet, visible and near- infrared region. The performance characteristics of interference filters are significantly superior to those of absorption (coloured) filters. The effective bandwidths of these filters are narrower than absorption filters.

2.4.3 Monochromators

As mentioned above, **monochromators are devices that can selectively provide radiation of a desired wavelength out of the range of wavelengths emitted by the source.** These are of two types; the prism and grating monochromators. These are described in the following paragraphs.

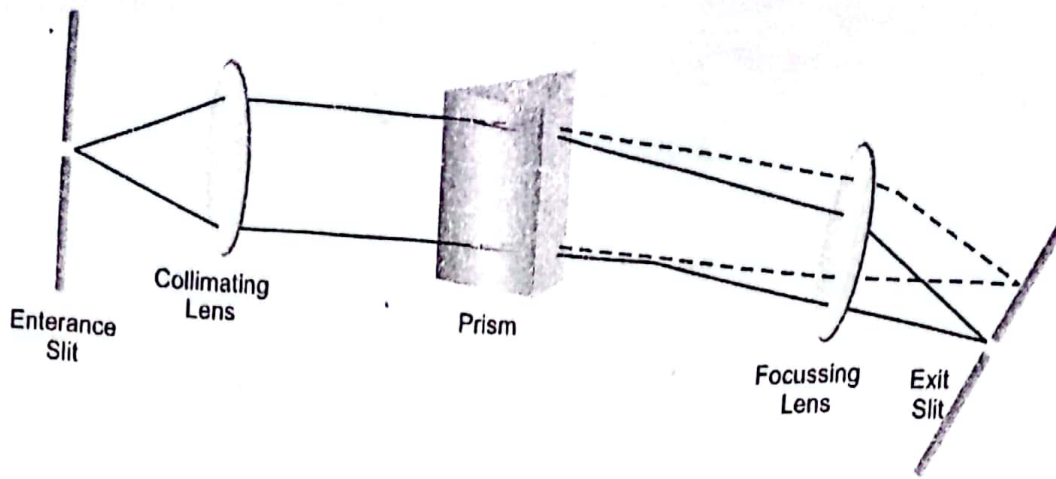
Prism Monochromators

You know that a **prism disperses** sunlight into seven different colours. This occurs due to the **refraction of the light** when it passes through the prism. The radiations of different colours having different wavelengths are refracted to different extent due to the difference in the refractive index of glass for different wavelengths. **Shorter wavelengths are refracted more than longer wavelengths** as depicted in Fig. 2.14 (a).



(a)

Fig. 2.14: a) Dispersion of radiation by prism



(b)

Fig. 2.14: b) Schematic diagram of the prism monochromator

If a prism is rotated, different wavelengths of the radiation, coming out after refracting through it, can be made to pass through the exit slit. In a **prism monochromator**, shown in Fig. 2.14 (b), a fine beam of the light from the source is obtained by passing through an entrance slit. This is then **collimated** on the prism with the help of a lens. The refracted beams are then focused on an exit slit. The prism is then rotated in a predetermined way to provide the desired wavelength from the exit slit.

Grating Monochromators

A grating is made by cutting or etching a series of closely spaced parallel grooves on the smooth reflective surface of a solid material as shown in Fig. 2.14 (a). The surface is made reflective **by making a thin film of aluminium on it** and the etching is done with the help of a **suitably shaped diamond tool**. The intensity of radiation reflected by a grating varies with the wavelength, the wavelength of maximum intensity being dependent on the angle from which the radiation is reflected from the surface of the line of the grating as shown in Fig. 2.15.

The gratings used for the ultraviolet and visible region generally contain about 1200-1400 grooves/mm.

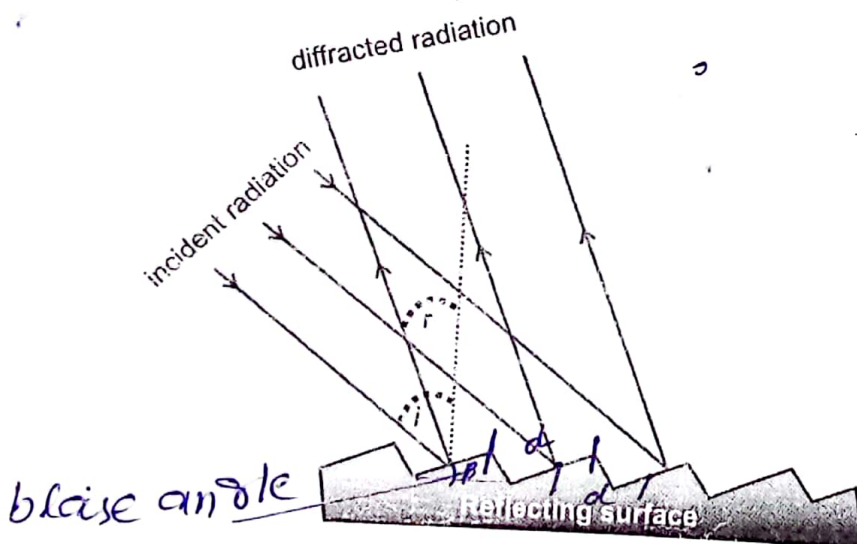


Fig. 2.15: Diffraction of Radiation by a grating

In grating monochromator (Fig. 2.16), a fine beam of the light from the source falls on a concave mirror through an entrance slit. This is then reflected on the grating which disperses it. The dispersed radiation is then directed to an exit slit. The range of wavelengths isolated by the monochromator is determined by the extent of dispersion by the grating and the width of the exit slit. Rotation of the grating in a predetermined way can be used to obtain the desired wavelength from the exit slit.

$$m\lambda = d \sin \theta$$

Not order (position)
λ changes

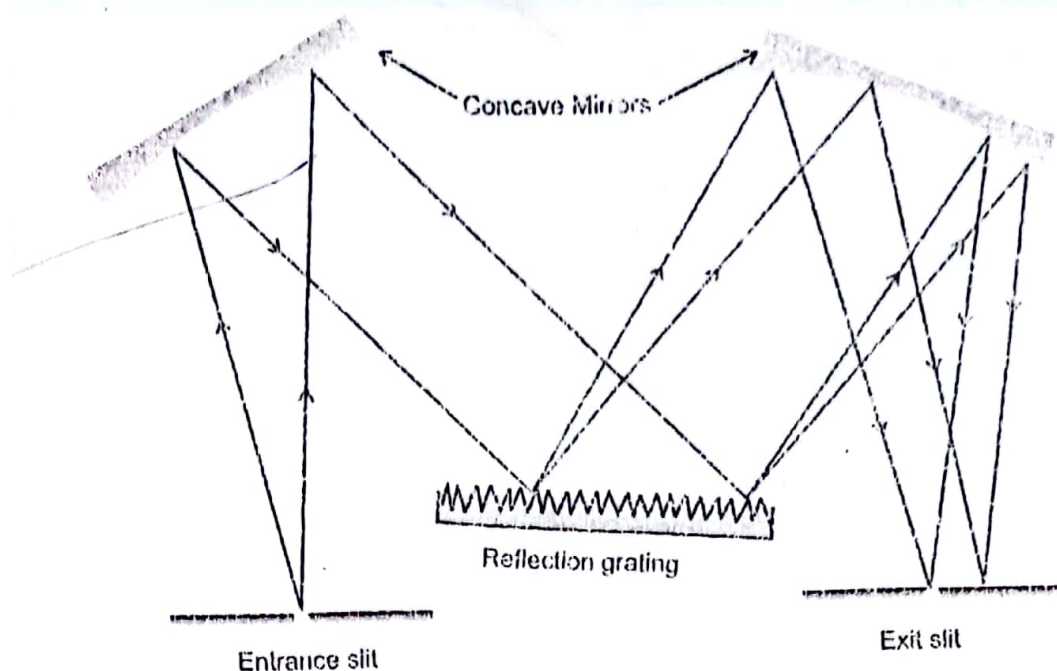


Fig. 2.16: Schematic diagram of a grating monochromator

Explain sample handling.

2.4.4 Sample Handling

The UV-VIS absorption spectra are usually determined either in vapour phase or in solution. In order to take the UV spectrum of the analyte it is taken in a cell called a cuvette which is transparent to the wavelength of light passing through it. A variety of quartz cuvettes are available for the spectral determination in the vapour phase. These are of varying path lengths and are equipped with inlet and outlets. For measurements on solutions in the visible region the cuvettes made of glass can also be used. However, since glass absorbs the ultraviolet radiations, these cannot be used in the UV region. Therefore, most of the spectrophotometers employ quartz cuvettes (Fig 2.17), as these can be used for both visible and UV region. Usually square cuvettes having internal path length 1.0 cm are used, though cuvettes of much smaller path lengths say of 0.1 mm or 0.05 mm are also available.

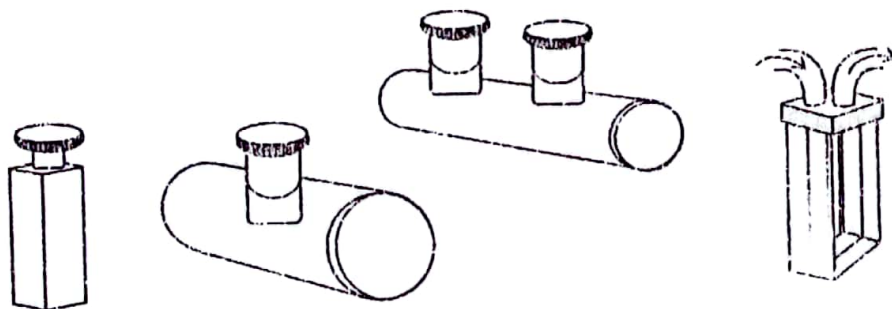


Fig. 2.17: Quartz cuvettes for measurements in solution and in vapour phase

The sample whose spectrum is to be measured is dissolved in a solvent that is transparent in the UV region. It means that it does not absorb in this region. Hexane, 95% ethanol, methanol and 1,4 dioxane are commonly employed as solvents. Now a days 'spectral grade' solvents are available which have high purity and have negligible absorption in the region of absorption by the chromophore.

In a typical measurement of a UV spectrum, the solution of the sample is taken in a suitable cuvette and the spectrum is run in the desired range of the wavelengths. The absorption by the solvent, if any, is compensated by running the spectrum for the solvent alone in the same or identical cuvette and subtracting it from the spectrum of the solution. This gives the spectrum only due to the absorption species under

investigation. In double beam spectrometers, the sample and the solvent are scanned simultaneously (subsection 2.5.2).

2.4.5 Detectors

The detectors are used to convert a light signal to an electrical signal which can be suitably measured and transformed into an output. The detectors used in most of the instruments generate a signal, which is linear in transmittance i.e. they respond linearly to radiant power falling on them. The transmittance values can be changed logarithmically into absorbance units by an electrical or mechanical arrangement in the signal read out device. There are three types of detectors which are used in modern spectrophotometers. These are described in the following paragraphs.

1. Phototube

Air

A phototube consists of a photoemissive cathode and an anode in an evacuated tube with a quartz window as shown in Fig. 2.18 (a). These two electrodes are subjected to high voltage (about 100 V) difference. When a photon enters the tube and strikes the cathode, an electron is ejected and is attracted to the anode resulting in a flow of current which can be amplified and measured. The response of the photoemissive material is wavelength dependent and different phototubes are available for different regions of the spectrum.

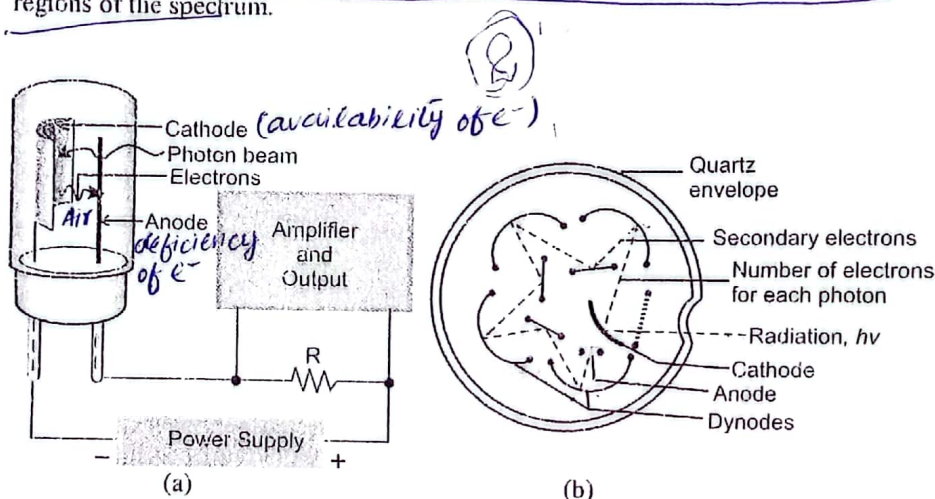


Fig. 2.18: Detectors of UV-VIS radiation; a) Phototube and b) Photomultiplier tube

2. Photomultiplier (PM) Tube

A photomultiplier tube (Fig. 2.18 (b)), consists of a series of electrodes, called **dynodes**. The voltage of successive electrodes is maintained 50 to 90 volt more positive than the previous one. When a radiation falls on the cathode an electron is emitted from it. This is accelerated towards the next photoemissive electrode by the potential difference between the two. Here, it releases many more secondary electrons. These, in turn are accelerated to the next electrode where each secondary electron releases more electrons. The process continuous upto about 10 stages of amplification. The final output of the photomultiplier tube gives a much larger signal than the photocell.

3. Diode Array Detector

These detectors employ a large number of silicon diodes arranged side by side on a single chip. When a UV-VIS radiation falls on the diode, its conductivity increases significantly. This increase in conductivity is proportional to the intensity of the radiation and can be readily measured. Since a large number of diodes can be arranged together, the intensity at a number of wavelengths can be measured simultaneously.

PP. L.A. (X) Aldehydes and Ketones $\Rightarrow$ Write four reactions with aldehyde or ketone to hydrocarbon.

Aldehydes and ketones can be reduced to
(a) H/C (b) Alcohol (c) 1,2-diols.

(a) To hydrocarbon

Question#(5)

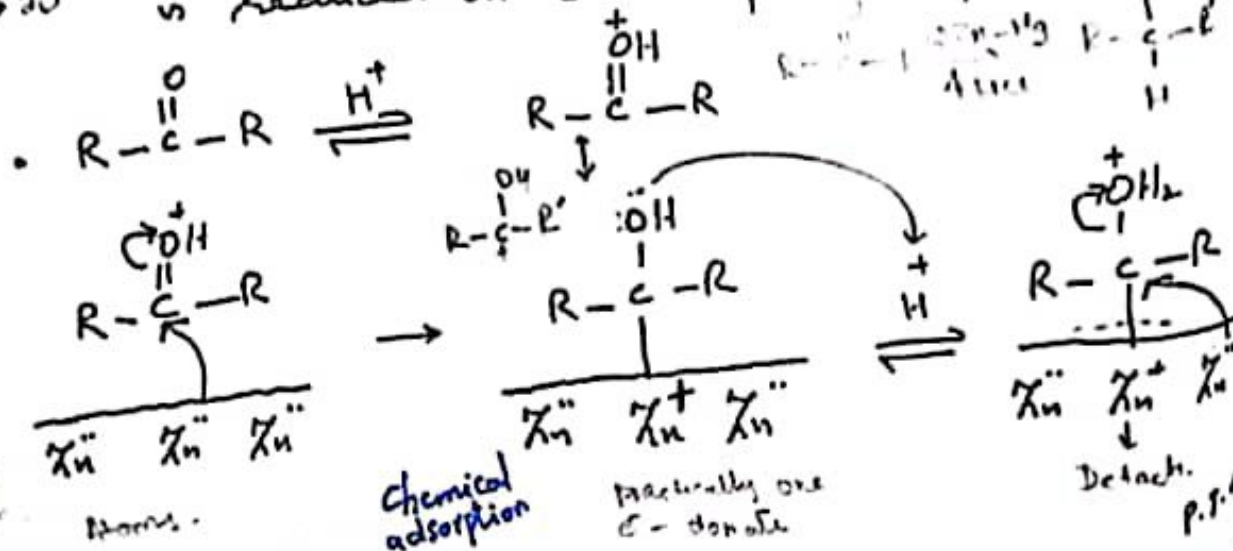
• Different methods are available based on the sensitivity of other functional centres in the reactant molecule in reducing condition.

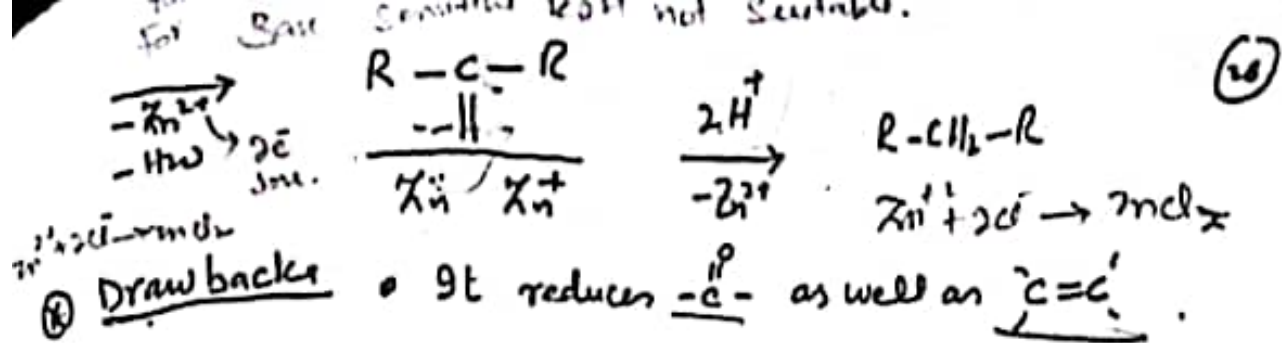
PP (ii) Clemmensen method: Better for ketones. Use to produce reactivity of Zn. $\uparrow$ Zn-Hg
Difference: In this reaction, we use zinc amalgam (Zn-Hg) in presence of conc. HCl.

Aldehyde
↓
primary alcohol
Ketone
↓
secondary alcohol.

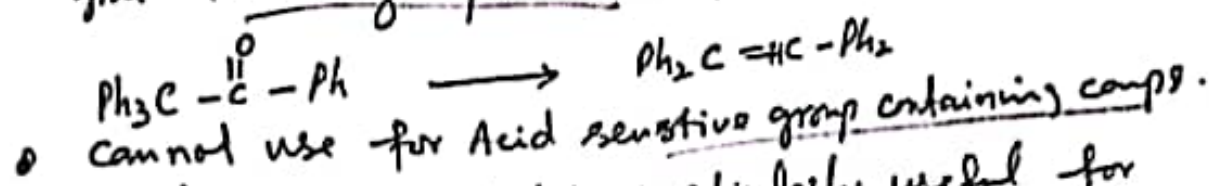
• The reaction is carried out often in the presence of toluene to produce three phase system.
• Most of ketone remain in upper toluene layer.
• The protonated ketone in the aqueous layer

211 → 20 is reduced on Zn surface as follow.

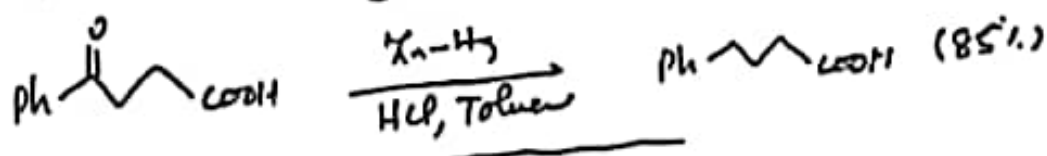




- ⊗ Drawbacks
- It reduces $\text{C}=\text{O}$ as well as $\text{C}=\text{C}$.
 - It also reduces Benzoyl alcohols and halides by hydrogenolysis.
 - Hindered ketones give low yield and sometimes give rearranged products e.g.

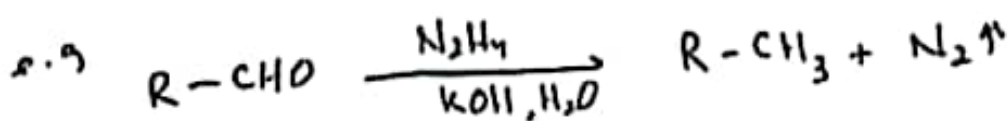


- ⊗ Merits This method is particularly useful for ketones having phenolic or carboxylic groups e.g.



(ii) Wolf-Kishner method

- Aldehydes are reduced to H/C by the treatment of hydrazine in presence of a strong base like KOH or NaOH.
- First hydrazone is formed and then it decomposes by strong base to give H/C.



the sample absorbance substance

S/Q Beer-Lambert law :: Above two laws can be merge to form a single law as

When a beam " The absorption of light by a substance at a particular wavelength is proportional to molar concentration as well as to path length of substance." / sample

Mathematically

Question#(6)

$$\log \frac{I_0}{I} = \epsilon \cdot c \cdot l \quad \text{--- (A)}$$

I_0 = Intensity of incident light

I = Intensity of transmitted light

c = Concentration in mol/dm³

l = path length or thickness of sample

ϵ = Proportionality constant known as molar absorptivity or molar extinction co-efficient

measure intensity where
if $I = I_0$
 $\epsilon = \frac{A}{c \cdot l}$
amount can be found out

The term $\log \frac{I_0}{I}$ is called optical density or absorbance. It is represented by A so
in any spectroscopy

$$\log \frac{I_0}{I} = A \quad \text{and equation (A) becomes}$$

$$A = \epsilon \cdot c \cdot l \quad \text{--- (B)}$$

Beer-Lambert's law

Beer's law and Lambert's law are usually combined and the combined law is known as Beer-Lambert's law. This law can be stated as "the rate of decrease of intensity of monochromatic beam of light with respect to thickness of the medium is directly proportional to the intensity of light and concentration of absorbing substance in solution".

Mathematically Beer-Lambert's law can be written as

$$\alpha \propto \text{M.C. of absorbing substance} \quad -\frac{dI}{dl} \propto I.C.$$

Where I is the intensity of light, C is the concentration of absorbing substance in solution and $-dI/dl$ is the rate of decrease of intensity of light with respect to thickness of medium or optical path length.

The above expression can be written as

$$\alpha \propto \text{path length of sample} \quad -\frac{dI}{dl} = K.I.C.$$

Where K is proportionality constant. Its value depends upon nature of absorbing material and wavelength of light used. Equation (10.2) is variable separable 1st order differential equation and it can be integrated by separating variables as

$$\int \frac{dI}{I} = -K.C \int dl$$

$$\ln I = -K.C.I + \text{constant} \quad (10.3)$$

The value of constant of integration can be determined using initial conditions i.e. when $l=0$ then $I=I_0$. By applying these conditions on equation (10.3), we get

$$\ln I_0 = \text{constant} \quad (10.4)$$

By putting the value of constant from equation (10.4) into equation (10.3), we get

$$\ln I = -K.C.I + \ln I_0$$

$$A = \epsilon \cdot C \cdot l$$

$$A = 0 \times \text{moldm}^{-3} \times \text{cm}$$

$$\ln 2.303 \log$$

$$\ln \left(\frac{I}{I_0} \right) = -K.C.I$$

$$\ln \left(\frac{I_0}{I} \right) = K.C.I$$

$$2.303 \log \left(\frac{I_0}{I} \right) = K.C.I$$

$$\log \left(\frac{I_0}{I} \right) = \frac{K}{2.303} \cdot C.I$$

$$\log \left(\frac{I_0}{I} \right) = \epsilon \cdot C.I$$

Absorptivity constant

experimentally can change.

$$(10.5)$$

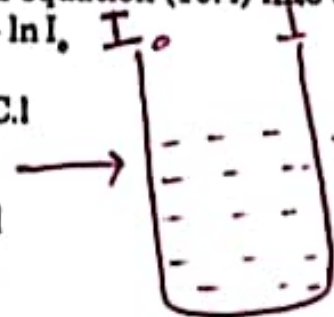
Where ϵ is equal to $K/2.303$. It is called molar extinction coefficient or molar absorption coefficient or molar absorptivity when concentration is expressed in moldm⁻³ and optical path length in cm. Equation (10.5) can be rearranged in exponential form as

Absorption :-

- Intensity of light
- Concentration of medium
- Path length

$$(10.2)$$

solution will be distorting case ✓



Intensity changes concentration remain same.

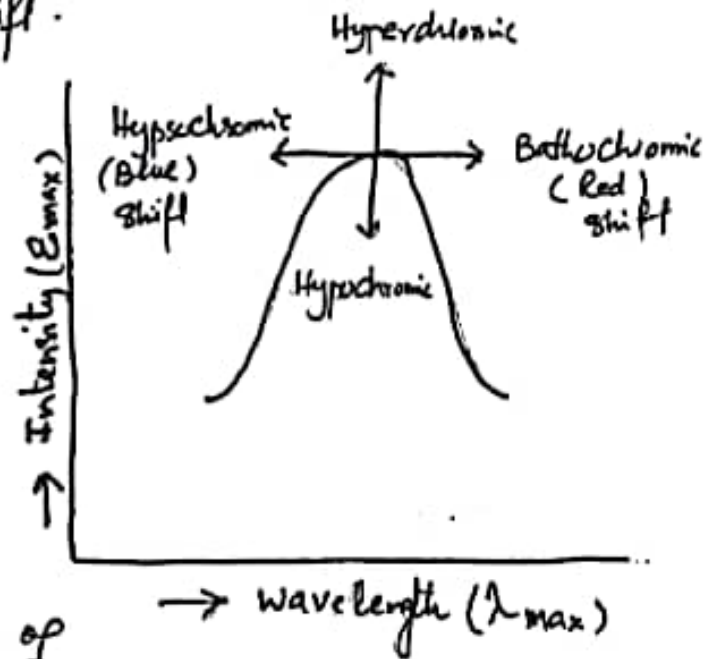
same.

Absorption & intensity shifts

(21)

- (a) Bathochromic shift: It is an effect by which absorption maximum is shifted towards longer wavelength due to presence of an auxochrome or change of solvent. It is also called Red shift.

e.g. in carbonyl compounds, decrease in solvent polarity shows red shift



- (b) Hypsochromic shift:

It is an effect in which absorption maximum is shifted toward shorter wavelength due to removal of conjugation or change of solvent.

It is also called blue shift.

e.g. aniline show absorption at 280 nm but in acidic solution it shows at 230 nm.

It is because in acidic solution nitrogen makes co-ordinate covalent bond and its lone pair is no longer available for conjugation of benzene.

(c) Hyperchromic effect :

- It is an effect by which the intensity of absorption maximum increases i.e. E_{max} increases.
- It usually happens when we introduce an auxochrome to compound.

e.g. for pyridine at 257 nm, E_{max} is 2750 is shifted to $\therefore 3560 (E_{max})$ at 262 nm for 2-methylpyridine.

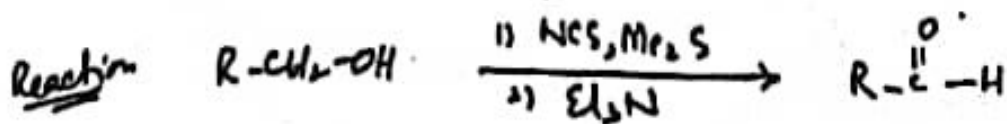
(d) HypoChromic effect : It is an effect by which intensity of absorption maximum decreases i.e. E_{max} decreases.

- It is usually due to the introduction of a group which distorts the geometry of molecule.

e.g. Biphenyl absorbs at 250 nm ($E_{max} = 19000$) but 2-methyl biphenyl absorbs at 237 nm ($E_{max} = 10250$)

(iv) Corey-Kim oxidation

- reagents: DMSO , NCS (N-chlorosuccinimide), Et_3N

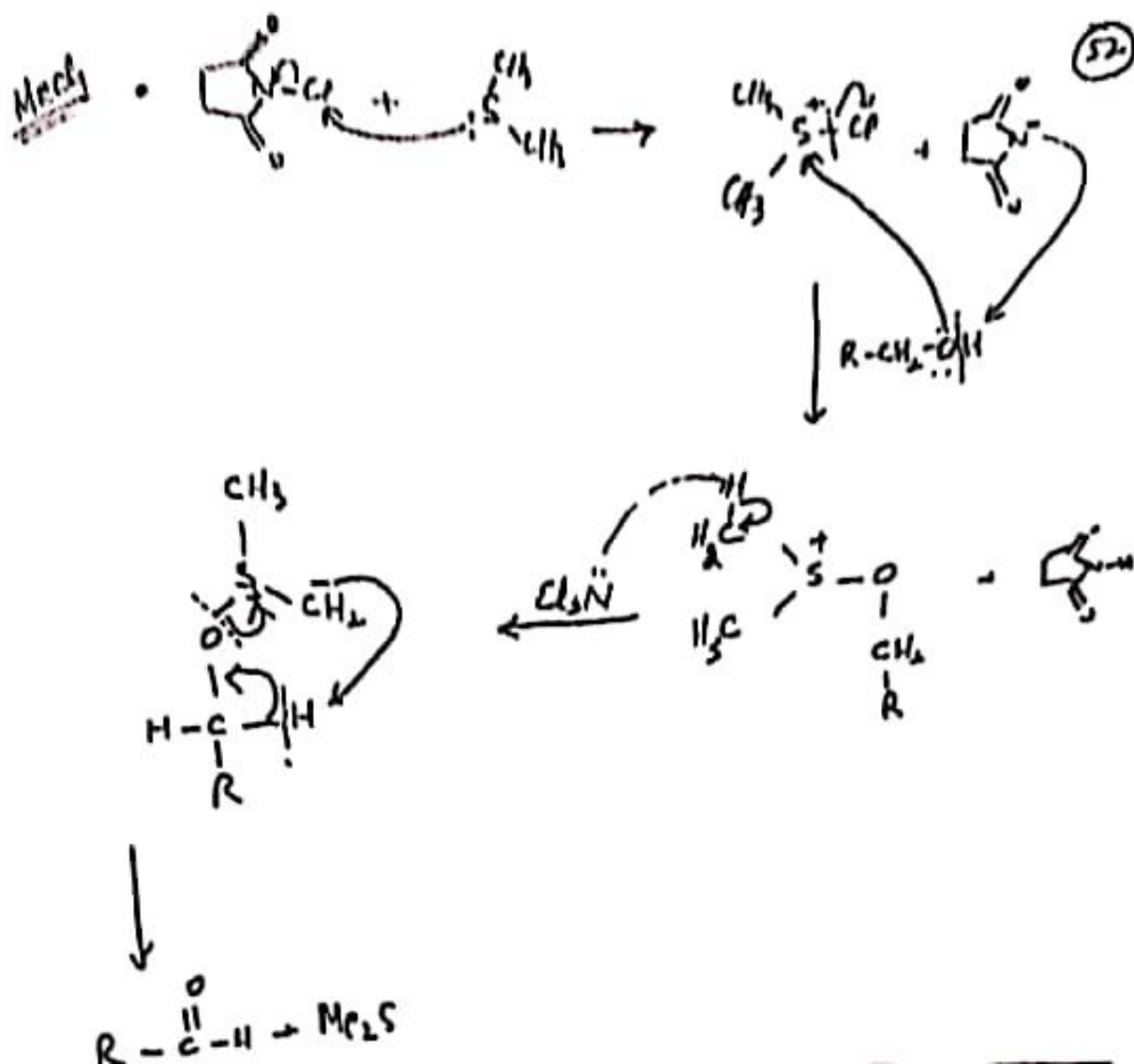


Long question#2

(1)

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Reductive Amination

[< Previous](#)



[Next >](#)

Long question2
(2)

Last updated: September 21st, 2022 |

Making Substituted Amines Through Reductive Amination

- ◆ Direct alkylation of amines with alkyl halides is a difficult reaction to control.
- ◆ One alternative that works extremely well is to form the **imine** of an amine with an appropriate aldehyde or ketone, and then to reduce the imine to an amine. This is called **reductive amination** and avoids the problem of multiple alkylations.
- ◆ One common reducing agent for this purposes is sodium cyanoborohydride (NaBH_3CN) which can selectively reduce imines in the presence of aldehydes. Many other reducing agents (NaBH_4 , $\text{NaBH}(\text{OAc})_3$, etc.) can be used as well.

- ◆ It's possible to start with a primary amine and do two successive reductive aminations, obtaining a tertiary amine.
- ◆ Reductive amination does not work for forming bonds from nitrogen to aromatic rings.

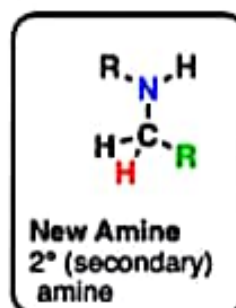
Summary: Reductive Amination

A versatile way of making different amines is to make an imine and then treat it with a **reducing agent** such as NaBH_4 (sodium borohydride) or NaCNBH_3 (sodium cyanoborohydride) to obtain the amine.

Step 1: Imine formation



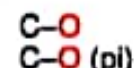
Step 2: Reduction



Key Bonds Formed



Key Bonds Broken



if desired this new amine can then be used in another reductive amination with a different aldehyde to make a 3° amine

- Reductive amination allows for amines to be "built up" sequentially from various aldehydes in a **controlled fashion** (as opposed to alkylation of amines with $\text{R}-\text{X}$ which tends to be hard to control)
- Using NH_3 with an aldehyde or ketone gives a **primary amine**
- Ketones can also be used to go directly from 1° amines to 3° amines
- Intramolecular reductive aminations give **cyclic amines**
- Useful reducing agents include NaCNBH_3 (sodium cyanoborohydride), $\text{NaH}(\text{OAc})_3$ (sodium tri-acetoxborohydride) and NaBH_4 (sodium borohydride)
- Does *not* work for making aromatic amines (i.e. $\text{Ph}-\text{NH}_2$)

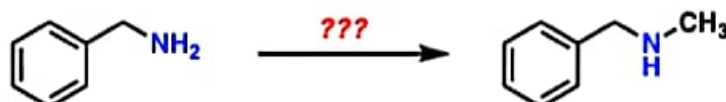
1. How Do We Make This Amine, If Direct Alkylation Doesn't Work?

Say you have a primary amine such as benzylamine and would like to make *N*-methylbenzylamine. How do you do it?

Synthesis of Amines

Say you're asked to make the following molecule (*N*-methylbenzylamine) from benzylamine.

How would you do it?



Direct treatment of benzylamine with an alkylating agent (e.g. methyl iodide, CH_3I) will result in significant formation of the undesired tertiary amine (i.e. *di*-alkylation).

Yes, you could try and separate out the secondary amine that's formed from the tertiary amine, but we're not going to settle for 10-30% yields here. Separating mixtures is fine on paper, but (*trust me on this*) it can be a real a pain in practice. Is there another way to do it?

One way that doesn't work well: direct alkylation



2. Reductive Amination To The Rescue

Enter reductive amination!

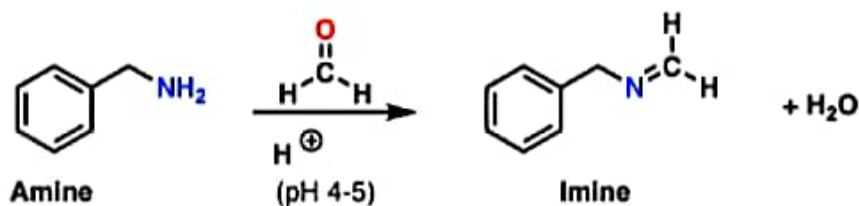
While alkylation can happen multiple times on an amine, **imines** only form once on a given amine. Once the imine is formed, the C=N bond can be reduced, giving us a new alkyl group attached to nitrogen.

This is a much more controlled manner of forming nitrogen-carbon bonds.

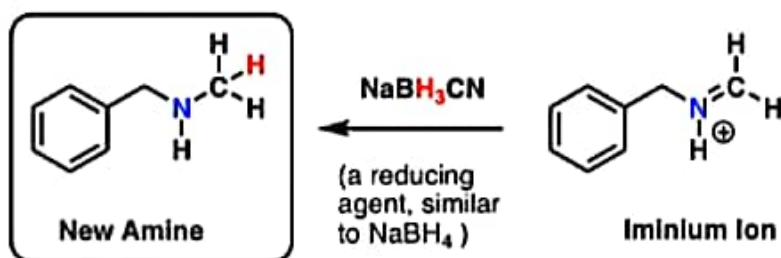
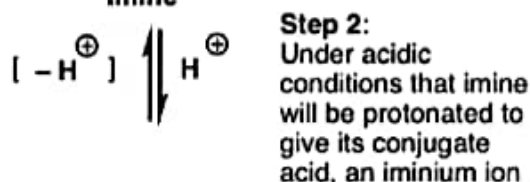
A Better Way To Do It: Reductive Amination

A more versatile way to form N-C bonds is to use a process called **reductive amination**. This involves transforming the amine into an imine, and then reducing the imine.

Step 1: Form an imine. We need to add a single carbon to our amine, so we add a carbonyl group with a single carbon - methanal (formaldehyde)



Step 3: In the presence of a reducing agent like NaBH_3CN the iminium will be reduced to give a new amine



We covered imine formation previously (*See article: [Imines – Properties, Reactions, Mechanisms](#)*). To refresh yourself on the mechanism, [hover here](#) and an image will pop up ([link](#)).

After the imine is formed, it must be reduced to the amine. It's possible to use the familiar reducing agent [sodium borohydride](#) (NaBH_4) for this process. You may recall that NaBH_4 is used for the reduction of aldehydes and ketones.

There are two other commonly used reductants for reductive amination: sodium cyanoborohydride (NaBH_3CN) and sodium tri-acetoxymethylborohydride ($\text{NaBH}(\text{OAc})_3$). For our purposes, they can be considered to be the same. In practice, NaBH_3CN is a little bit better than NaBH_4 .

[For a discussion as to why NaBH_3CN tends to be a better choice than NaBH_4 , see [Note 2](#). In the schemes below, we'll use NaBH_3CN , but NaBH_4 and $\text{NaBH}(\text{OAc})_3$ can be considered to work just as well. 1

3. Applying Reductive Amination

Reductive amination is extremely versatile and can be used to install a large variety of different alkyl groups on an amine. The nice part is that the groups just go on once.

The table below shows examples of how methyl, ethyl, propyl, butyl, and benzyl groups can be installed on an amine, by employing an appropriate aldehyde.

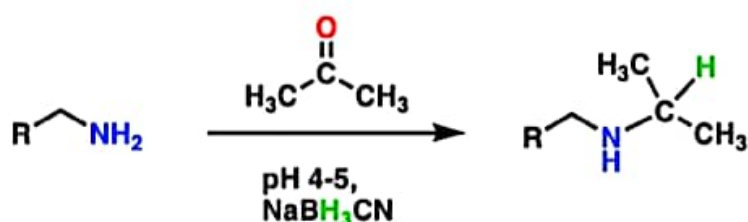
(Note that reductive amination doesn't work for installing a phenyl group. Why not? Note 1.)

Desired group to attach to nitrogen	Aldehyde to use	Example of reaction
Methyl (CH ₃)		
Ethyl (CH ₂ CH ₃)		
Propyl (CH ₂ CH ₂ CH ₃)		
Benzyl (CH ₂ Ph)		
Phenyl	—	<i>can't be done! (use Buchwald-Hartwig instead)</i>

4. Ketones Also Work

What about ketones? They work too!
Employing a ketone will result in a branched alkyl substituent on the amine.
For example, using acetone in the following reductive amination gives an isopropyl group.

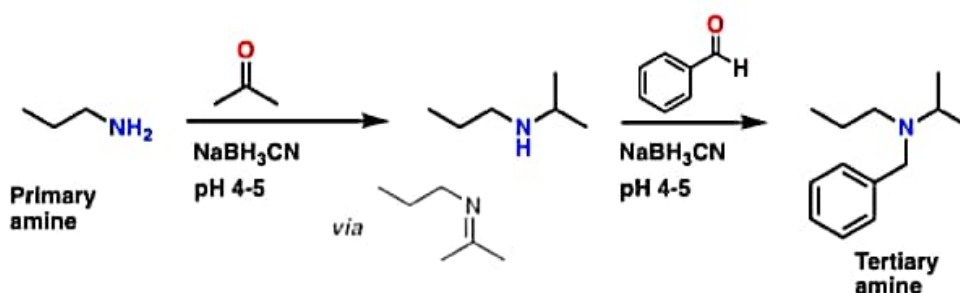
Can also be used with ketones



5. Sequential Reductive Aminations

Another useful feature of reductive amination reactions is that two (or three if one starts with ammonia) reductive aminations can be employed in sequence. For example, look at the synthesis of the tertiary amine below.

Two successive reductive aminations provide a tertiary amine from a primary amine

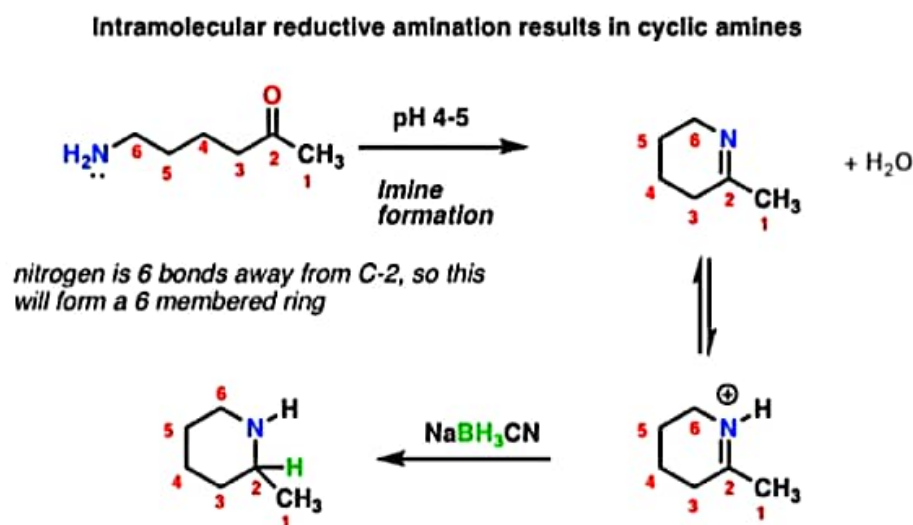


(note: the synthesis could have proceeded just as well had the order of the two reductive aminations been reversed)

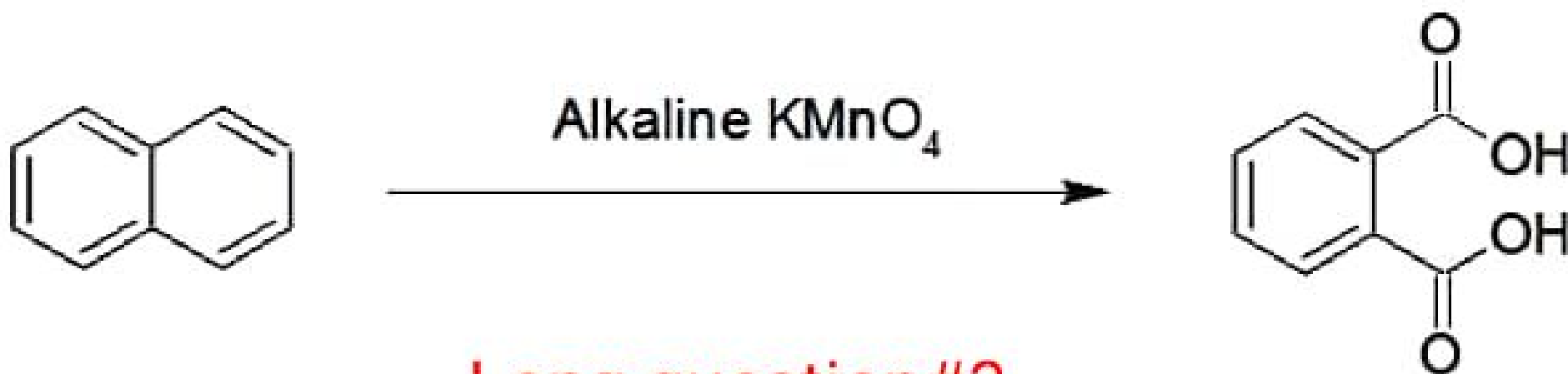
Importantly, the sequence of reactions isn't crucial here. We could have done the first reductive amination with benzaldehyde first, and acetone second, and still obtained the same product.

6. Intramolecular Reductive Aminations

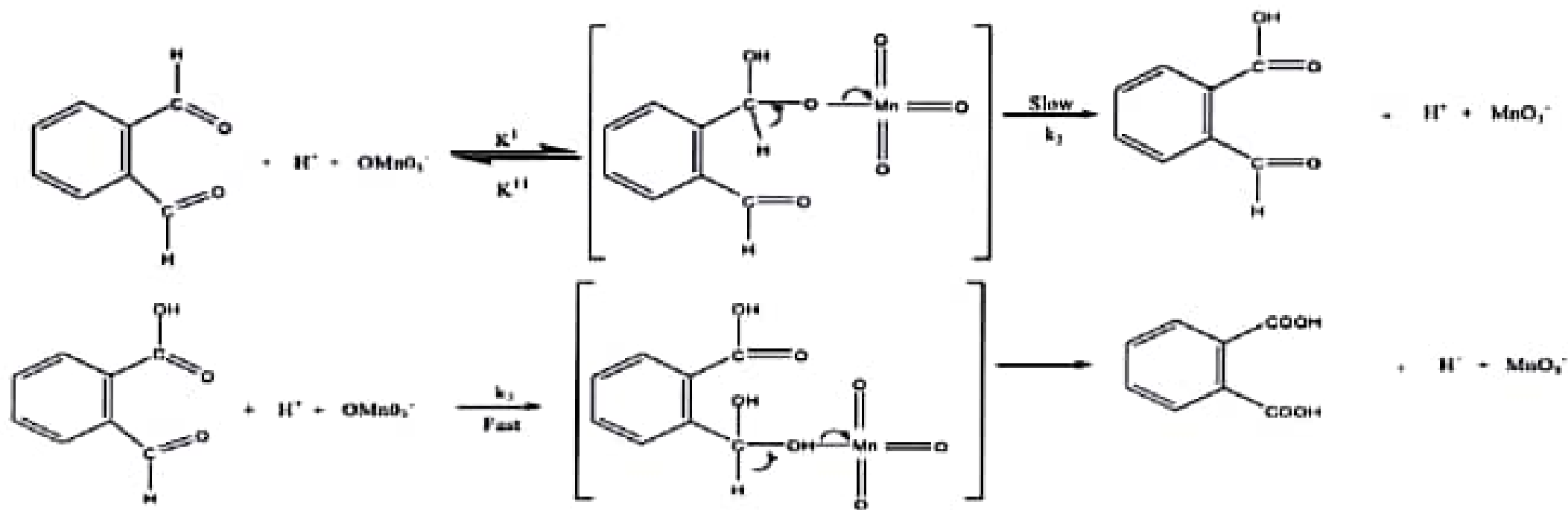
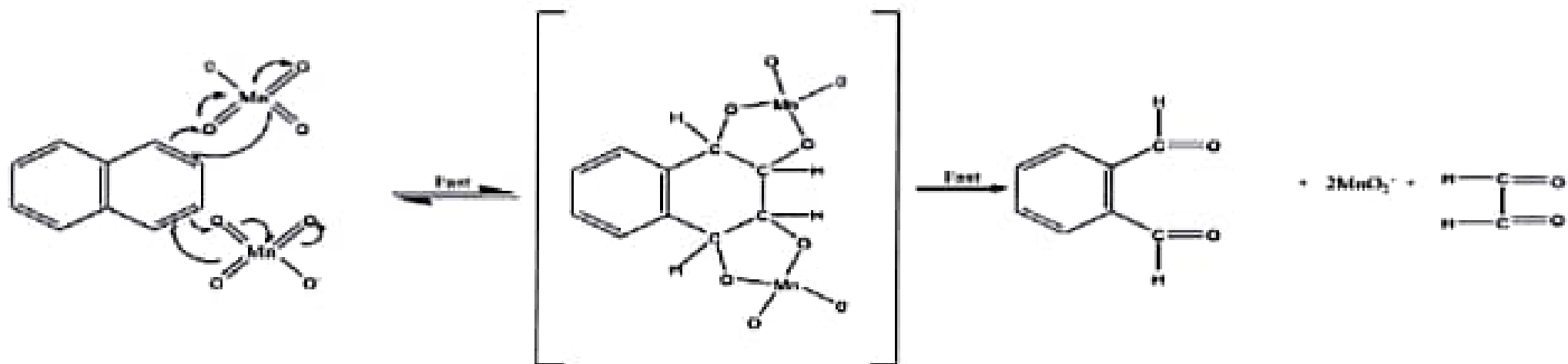
Finally, there's the intramolecular case, which always seems to give students a headache. If a molecule contains both an amine and a carbonyl, then it can cyclize to give a cyclic amine.



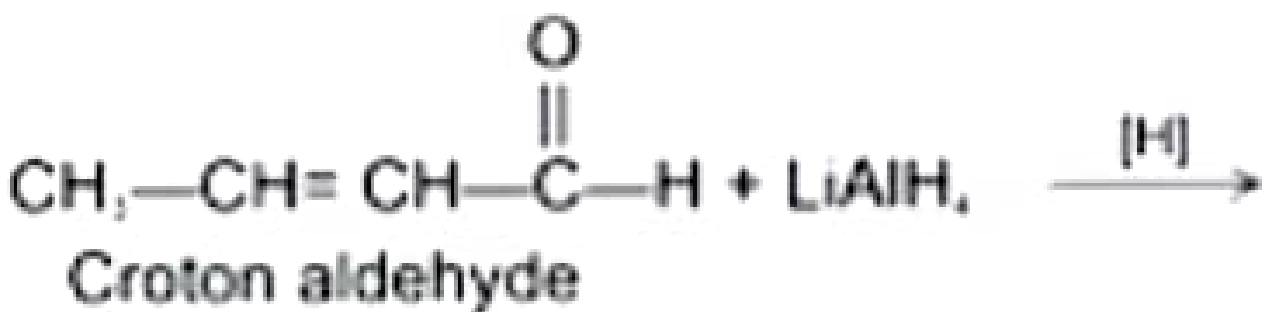
When drawing out the product of ring formation, I strongly advise counting and numbering your carbons. I've seen so many students make mistakes when redrawing that it's well worth the time to double check that you haven't left anything out



Long question#3



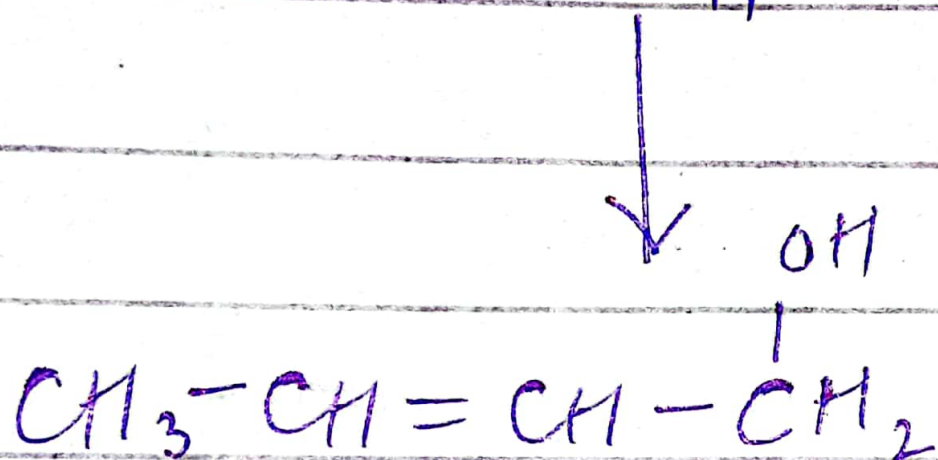
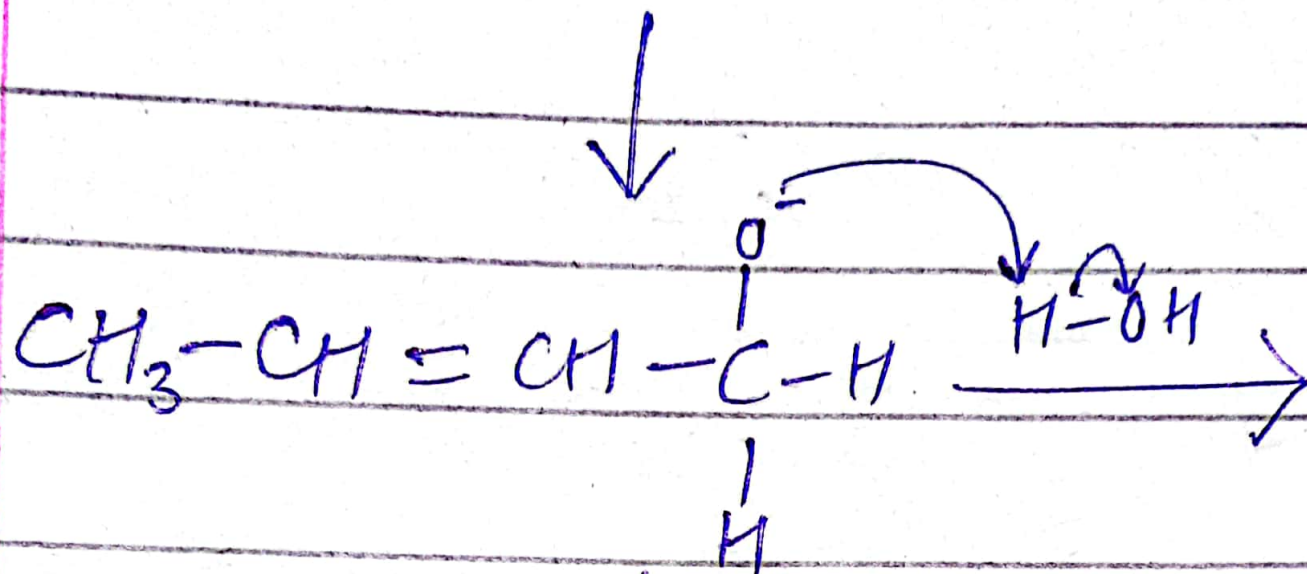
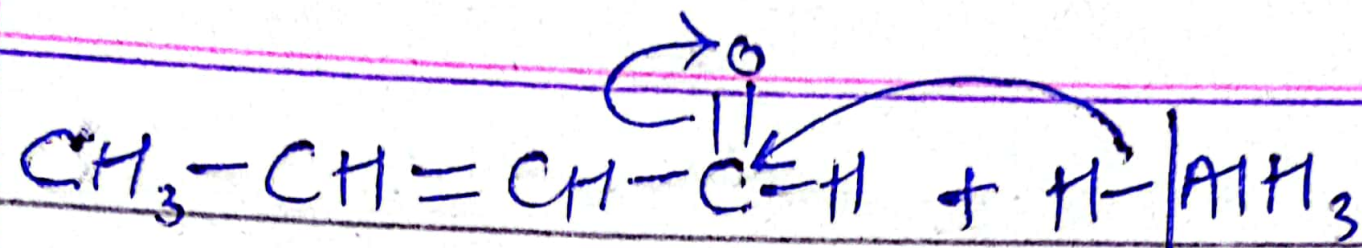
If LiAlH_4 is not present in excess.



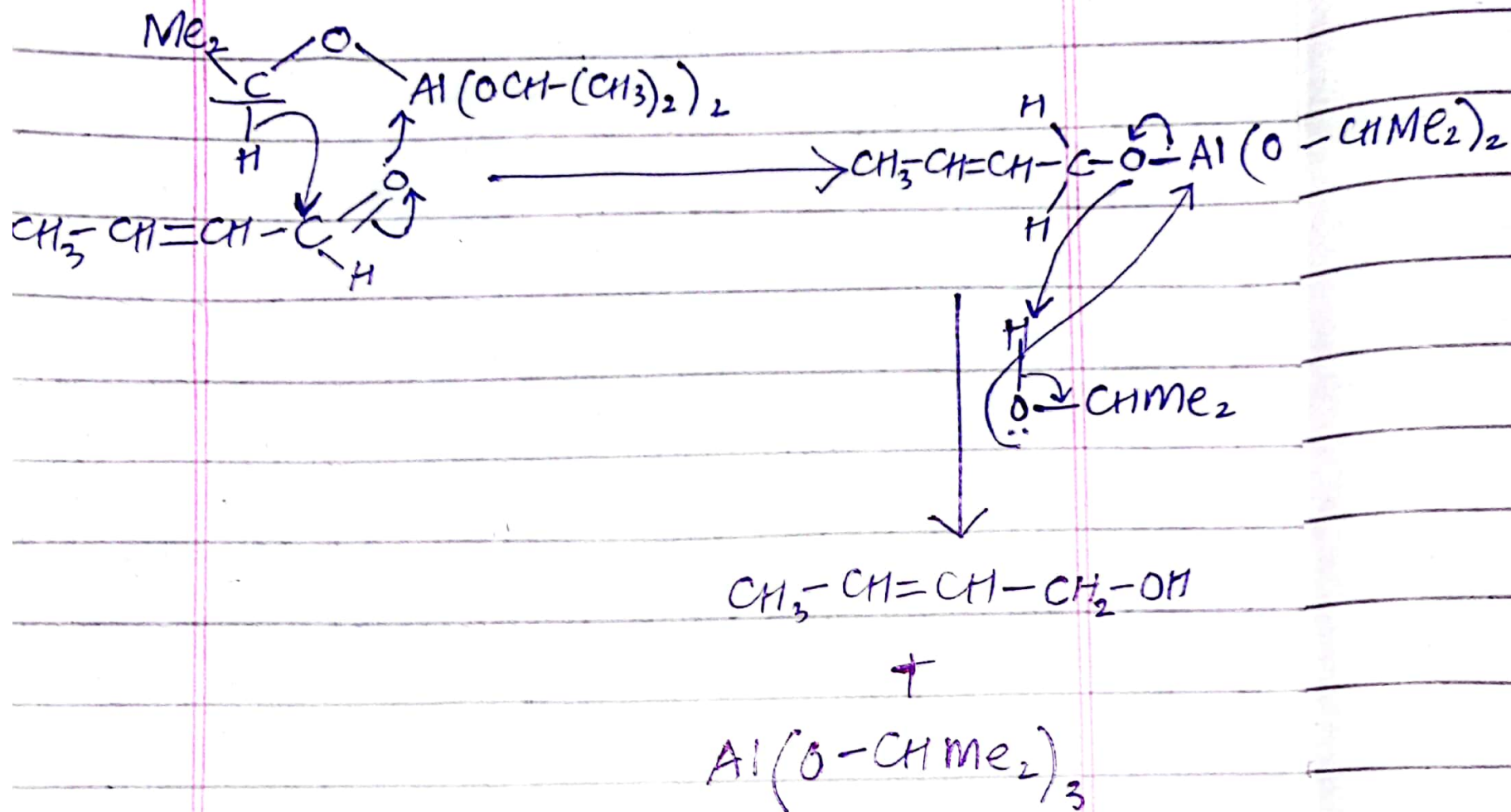
It will selectively reduce the carbonyl group. However in case of



(Both double bond and carbonyl group are reduced)



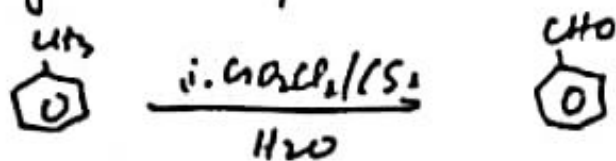
Meerwein-Ponndorf-Varley Reaction



- (i) Oxidation to aldehyde
- These methods are available for selective oxidation of ArCH_3 to ArCHO .

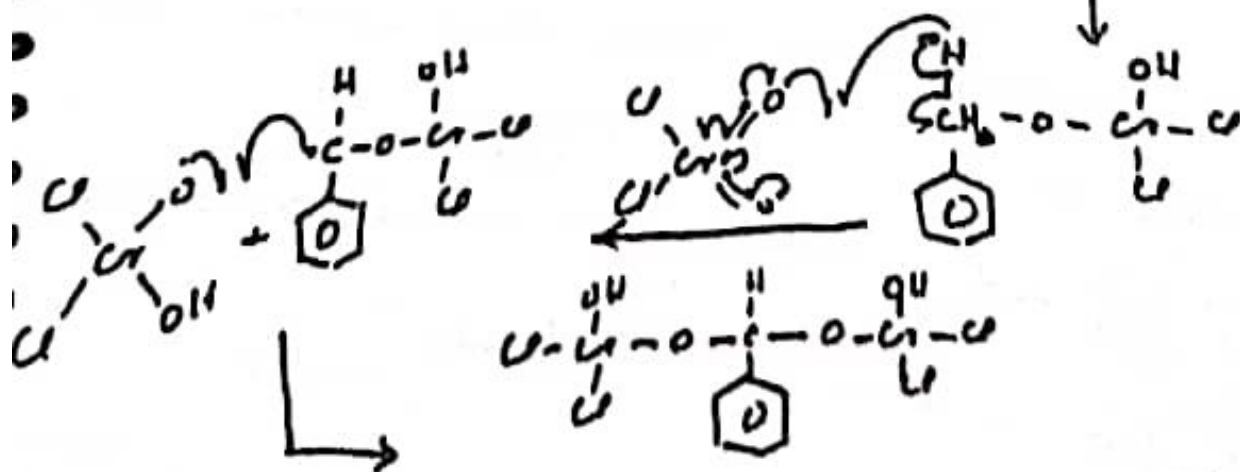
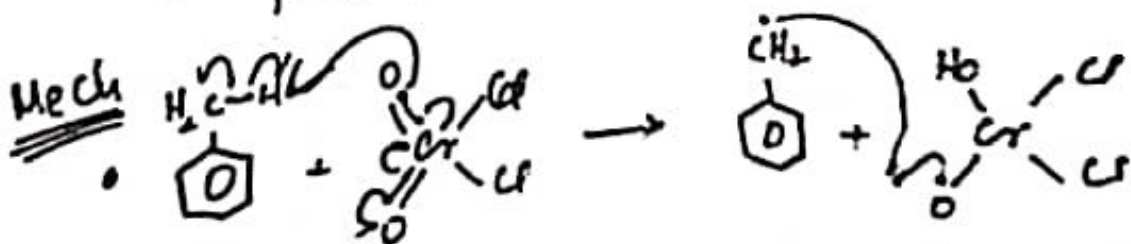
(ii) Etard Reaction (with CrO_2Cl_2)

- A solution of CrO_2Cl_2 in CS_2 and added dropwise to benzylic compound at $25-45^\circ\text{C}$ and brown complex is decomposed by H_2O to give aldehyde and chromic acid



Long question #4

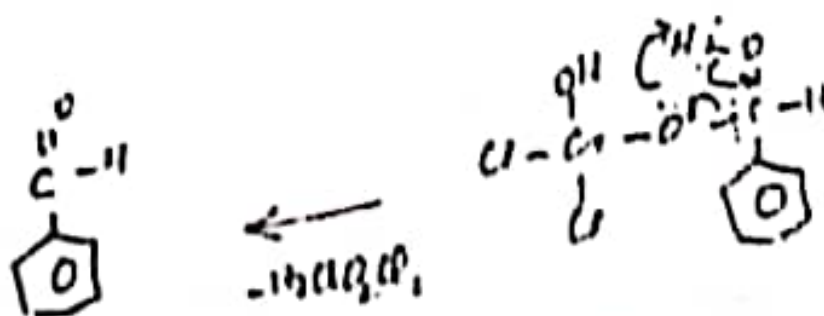
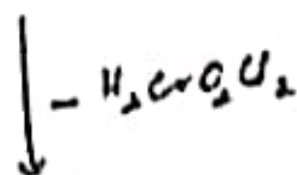
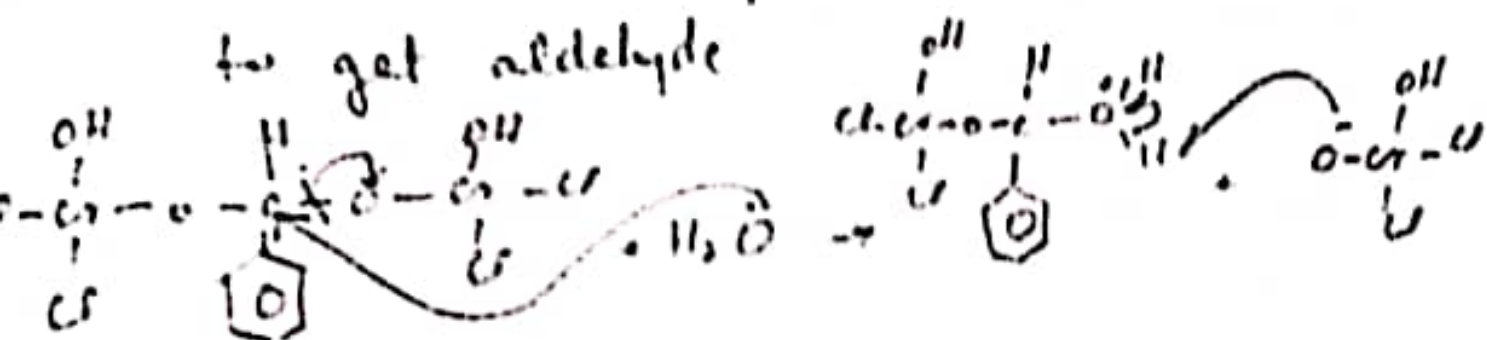
- Precaution :: The aldehyde formed must be rapidly removed by distillation or extraction to prevent its further oxidation



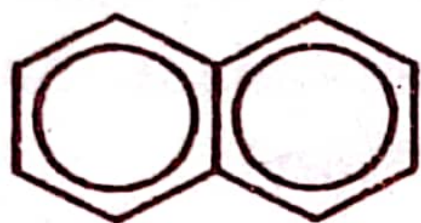
Brown ppt (complex)

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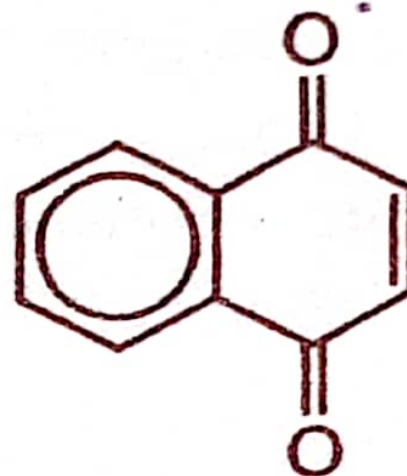
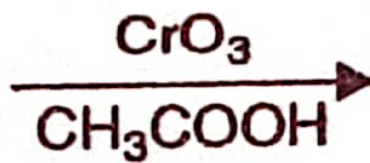
- The brown ppt. separated and H_2O is added (14) to get aldehyde



... mix Oxide in acetic anhydride.



Naphthalene



1,4-Naphthaquinone